

Research Article

Construction of Knowledge Graph for Emergency Resources

Heng Mu ¹, Peng Wu ², and Wenyi Su ¹

¹*School of Economics and Management, Nanjing University of Science and Technology, Nanjing 210094, China*

²*School of Intelligent Manufacturing, Nanjing University of Science and Technology, Nanjing 210094, China*

Correspondence should be addressed to Peng Wu; wupeng@njust.edu.cn

Received 3 July 2023; Revised 1 November 2023; Accepted 22 January 2024; Published 12 February 2024

Academic Editor: Costa Gianni

Copyright © 2024 Heng Mu et al. This is an open access article distributed under the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.

Knowledge graphs can effectively organize and represent information related to emergency resources for unforeseen sudden events. In this study, we construct a model layer for the knowledge graph of emergency resources, focused on sudden events, through the classification and analysis of unforeseen disaster measures. This study defines eight interconnected entity types, each characterised by a set of attributes and engaging in one or more relationships with other entity types. Utilizing 121 incident investigation reports from the emergency management departments of various provinces and cities over the past five years, we select five entities with the highest frequency of occurrence along with their corresponding four relationships. We then design an extraction plan for these entities and relationships. Based on the completed knowledge graph data, we formulate 14 questions related to emergency resources for sudden events and construct 19 corresponding question-and-answer templates using a template-based question-answering (QA) approach. We retrieve the corresponding Cypher statement templates through template mapping and obtain the question answers through querying. Finally, we design a knowledge graph question-and-answer system using the Django web framework, which includes entity queries and knowledge QA functions, specifically for emergency resources related to sudden events.

1. Introduction

Emergencies constantly threaten the safety and development of human society and the economy, as demonstrated by the shocking “9/11” terrorist attacks in the United States and the ongoing COVID-19 pandemic, both of which have had tremendous impacts on the world. The COVID-19 pandemic, which has been affecting the world since the end of 2019, has led to derivative events such as illegal immigration to evade the epidemic and the importation of food with falsified nucleic acid test reports. The pandemic has significantly impacted industries such as tourism, catering, transportation, and commerce, posing not only a safety hazard to public health but also causing substantial economic repercussions.

Appropriately allocating emergency resources can minimize losses during emergencies. The importance of these resources is emphasized in two major chapters of the “People’s Republic of China Emergency Response Law” under the headings of “Prevention and Emergency

Preparation” and “Emergency Response and Rescue.” However, China’s emergency resources currently face issues such as inadequate investment, a low degree of resource integration, and underperformance in management. The “14th Five-Year Plan for National Emergency System” reports that China’s emergency management foundation is weak and the level of scientific and information technology application is generally low. Early risk perception, identification, and warning are lacking, and there is insufficient assurance for emergency material supplies. Other significant problems include weak grassroots emergency response capabilities and inadequate public awareness of risk prevention. A significant gap exists between the capacity of emergency management systems and the requirements of modern national governance.

While most existing research on emergency resources focuses on the evaluation index system, the scope of emergency resource management is broader. Analyzing emergency resources without considering the overall picture can lead to biased results that are challenging to apply.

Constructing a comprehensive knowledge graph of emergency resources for sudden events can leverage the advantages of knowledge graphs. This approach integrates disordered and fragmented data into one map, thereby achieving “a map for the whole domain,” enabling rapid acquisition of information on necessary emergency resources in response to sudden events during emergency command. From a practical perspective, China’s emergency resource management is not well-coordinated or unified. Most emergency resources remain dispersed, leading to inconsistent and often insufficient resource data. This state of affairs also hampers research on emergency resources for sudden events. By categorizing typical sudden events, we can sort and guarantee the tasks for emergency resources that correspond to them. We can also integrate relevant information about the quantity, source organization, and effectiveness of various emergency resource configurations. This approach not only enables rapid access to essential information on emergency resources for corresponding typical sudden events but also provides a data foundation for other research and applications in the field of emergency resources for sudden events.

2. Relevant Research

2.1. Emergency Resource Research. Research on emergency resources has achieved considerable results by focusing primarily on the classification, layout, and dispatch of emergency resources. Zhou [1] proposed a framework for maritime emergency resources based on multiple criteria decision-making, geospatial techniques, and game theory. Chen et al. [2] proposed a new emergency dispatching framework for distribution systems from the perspective of the cyber-physical-social system. Under this framework, a novel cooperative scheduling model is also proposed, which aims to minimize time, cost, and shedding load as a mixed-integer nonlinear optimization model. Du et al. [3] proposed a mathematical model for emergency resource classification and presetting according to the distribution characteristics of demand points in a particular chemical park. The proposed model gives the optimal decision for the predeployment of emergency resources, and greatly reduces the total cost. The application of this model shows its applicability. Ma et al. [4] proposed a novel optimization model of maritime emergency resource allocation. By using the long short-term memory (LSTM) model to predict the number of possible accidents, a reasonable allocation strategy was formulated to meet the requirements of the actual situation, and the budget utilization rate reached 87.9%. Meng [5] designed an integrated social emergency resource monitoring system to maximize the number of resources collected and minimize logistics costs. Chen et al. [6] proposed a three-level network for cross-regional emergency resource scheduling and used an ant colony optimization design to build a cross-regional emergency resource scheduling model. Liao et al. [7] applied an artificial neural network (ANN) to develop an environmental emergency decision support system, and the results show the effectiveness of the proposed method. Yin et al. [8] designed

a preference representation method for large group emergency decision-makers based on the signal detection theory and constructed a risk measurement model using the rules of group preference fusion. Chen et al. [9] applied the knowledge graph to the emergency task and established the knowledge graph of the emergency case of a long-distance oil and gas pipeline for the first time. The results show the accuracy of the proposed model in urgent task recommendations. Previous studies neglected the effective representation of complex relationships by a knowledge graph. The introduction of the knowledge graph effectively solves the problem of unclear input of case features and shows a good practical application effect in emergency tasks.

2.2. Knowledge Graph Research. The knowledge graph [10] was proposed by Google in May 2012 to improve the quality and experience of user’s searches. Domain-specific knowledge graphs are used in military, public security, transportation, healthcare, and other fields for complex application analysis or decision-making assistance. They have the characteristics of high expert participation, complex knowledge structure, high knowledge quality requirements, and fine knowledge granularity. In the field of emergency, Xiao et al. [11] applied the knowledge graph to power grid fault handling and effectively improved emergency handling ability and dispatching intelligence level by extracting, expressing, and managing fault handling information. Through the proposed top-down and bottom-up construction methods of the knowledge graph, fault information analysis, intelligent decision-making, and multidimensional human-computer interaction are realized. Shin et al. [12] designed and constructed ontologies and knowledge graphs for 1,001 major chemical substances. They queried and reasoned candidate substances through the knowledge graph to enable rapid diagnosis and subsequent early response to chemical exposures at accident sites. Mao et al. [13] applies knowledge graphs to process safety in the chemical industry, proposing a comprehensive semi-automatic construction framework that includes ontology definition, data acquisition, import, and storage. It is found that the process safety knowledge graph can enhance the ability of knowledge-based analysis to discover hidden relationships between possible risk causes and consequences in emergencies and provide a basis for process safety-related applications.

In the construction of a knowledge graph, knowledge extraction is the key step. In recent years, with the advances in deep learning technology, numerous scholars have applied deep learning to entity extraction, whereby word vectors are used as the system input to achieve entity recognition through neural networks. Currently, the primary neural networks used for entity recognition are recurrent neural networks [14] and neural networks with attention mechanisms [15]. The main task of relationship extraction is to model text corpora to obtain semantic relationships between entities [16], thereby forming a network-like knowledge structure. There are also numerous methods for relationship extraction, such as template-based, machine

learning, deep learning, and open-domain methods. Relationship extraction methods based on deep learning can be subdivided into two categories as follows: pipeline methods and entity relationship joint extraction methods. The pipeline method separates entity and relationship recognition into two unrelated processes, with the effect of relationship recognition depending on the quality of entity recognition. Guo et al. [17] proposed a new attention and region-based convolutional neural network (Att-RCNN) model for text feature extraction and relationship classification, using attention mechanisms to strengthen keywords and features for improved classification performance. Gan et al. [18] proposed an entity attention LSTM network at the subsequence level to focus on the important information between two entities. The attention mechanism was introduced at the feature fusion stage to improve the context information processing ability of the model. Katiyar and Cardie [19] were the first to introduce the attention mechanism into bidirectional BiLSTM for joint entity relation extraction. The entity recognition task was completed using BiLSTM, and the relationship classification task accepted the entity sequence and encoding representation output of the entity recognition task as inputs. An attention model was introduced to effectively improve the shortcomings in the literature, such as relying on part-of-speech tags. Liu et al. [20] proposed a joint classification model based on multilabel classification. The model consists of a BiLSTM+DCNN encoder and a decoder for entity, relationship, and joint entity relationship prediction. They transformed the relationship extraction task into a multilabel classification task, thereby effectively solving the problem of overlapping entity relationships.

Knowledge question-answering (QA) systems are an important application of knowledge graphs. Knowledge graphs typically serve as knowledge bases, providing data support for QA systems. When users ask questions in natural language, the QA system provides concise and accurate answers [21]. The earliest intelligent QA can be traced back to the advent of artificial intelligence in the 1960s based on Alan Turing's famous Turing test [22]. The ability of machines to correctly answer questions is monitored to verify whether machines have intelligence. In recent years, the emergence of knowledge bases and knowledge graphs has integrated data and relationships into a structured system, thereby optimizing the quality of data. Concurrently, the rapid development of deep learning has made natural language processing very convenient. Intelligent QA systems have evolved from document-based to knowledge graph-based systems, which present a qualitative leap. Numerous studies have been conducted on knowledge graph-based QA systems in different domains, and relevant research has emerged on commensurate application methods [23–26]. The construction of knowledge-based QA systems can be generally divided into three categories as follows: template matching, semantic parsing, and deep learning. Deep learning-based methods use an end-to-end model to translate the question into a corresponding semantic representation sequence. Typically, deep learning algorithms map entities as well as relationships between entities to

a vector space and obtain answers by calculating the similarity between vectors.

Unlike previous studies, our work focuses on a specialized knowledge graph that is designed for sudden events, enhancing the entity types and relationships for a more accurate representation. By using advanced models, we overcome the limitations of traditional databases. Furthermore, we introduce a novel knowledge question-and-answer system, demonstrating a unique approach to emergency resource management and knowledge graph applications.

3. Construction of the Emergency Resource Knowledge Graph for Unexpected Incidents

3.1. Demand Analysis. In the event of an unexpected disaster, emergency resources are a primary concern. However, most people have insufficient knowledge about these resources. Consequently, both before and after the occurrence of an unexpected disaster, considerable resources (such as emergency reserves and assistance) are wasted, and the development of national society faces substantial risks. Relevant departments may struggle to estimate the number of various emergency resources required during an unexpected mishap, leading to untimely emergency responses. Therefore, this study targets relevant government emergency command departments in the construction of a knowledge graph to ensure the accuracy and completeness of emergency resource information for unexpected disasters. A knowledge graph can help these departments improve the detailed information available on emergency resources from previous unexpected disasters. Consequently, in subsequent unexpected disasters, emergency resources can be allocated more efficiently based on knowledge graph data. Meanwhile, the knowledge QA system, an application of the knowledge graph, primarily targets users who need information about emergency resources in the event of an unexpected disaster. Users seeking knowledge about unexpected emergency resources can ask questions based on their needs through the knowledge QA function and obtain responses from the system. The materials and methods section should contain sufficient detail so that all procedures can be repeated. It may be divided into headed subsections if several methods are described.

The framework of our work consists of the following steps: first, we analyzed the existing classification of emergencies and emergency resources and designed a suitable classification for the data of emergency investigation reports. We constructed a knowledge graph schema layer and defined the entities, relationships, and attributes of the emergency resource knowledge graph. Second, we cleaned the data of emergency investigation reports and other sources from government websites and performed entity extraction and relationship extraction. We stored the extracted data in a graph database. Third, we designed relevant questions and query functions based on the data characteristics of the investigation reports and used a Python web framework to build an emergency resource question-and-answer system for emergencies.

3.2. Classification of Emergency Resources for Sudden Mishaps. Classification of emergency resources for sudden disasters is primarily conducted in the schema layer. A logically consistent classification of emergency resources can improve the coherence and structure of the knowledge graph. There are numerous methods available for classifying emergency resources; therefore, selecting an appropriate method is crucial. Necessary adjustments should be made to the knowledge graph based on specific requirements and data before proceeding with the design of the schema layer.

Currently, the classification of sudden emergencies in China is well-established and primarily divided into the following four disaster categories: natural, accident, public health, and social security events. Natural disasters encompass phenomena such as floods, droughts, forest fires and meteorological, geological, marine, and biological disasters. Accident disasters include incidents such as coal mine and construction accidents, as well as road and water traffic accidents. Public health events cover infectious diseases, food and drug safety issues, and animal epidemic events. Social security events involve mass incidents, major criminal events, terrorist attacks, and cybersecurity incidents.

According to the Emergency Response Law, sudden emergencies are classified into four levels based on their severity and urgency as follows: exceptionally severe (level I), severe (level II), significant (level III), and general (level IV). The types of required emergency materials are diverse, as different emergencies necessitate different types of rescue materials. For effective management and rational allocation of emergency resources, these resources must be classified scientifically and meticulously according to (1) the urgency of emergency material usage, with three classification categories, i.e., general, severe, and urgent; (2) the purpose of emergency materials, with thirteen categories based on the Catalog of Classification and Product Directory of Emergency Materials; (3) the reasons for emergency material demand, with five classification categories, i.e., natural disaster, accident disaster, public health event, social security event, and economic security event; and (4) the scope of emergency material usage, divided into general and specialized categories.

Based on the classification statistics of emergency resources and the analysis of information from investigation reports on sudden events, this study classifies emergency resources into 16 categories. These include lighting equipment, protective equipment, rescue transport, engineering equipment, communication and broadcasting equipment, life rescue materials, temporary accommodation and food supplies, engineering materials, vehicles, life support items, pollution clean-up supplies, equipment tools, and power fuel. The abovementioned 13 categories of materials, along with human resources, financial resources, and mixed resources, make up a comprehensive emergency resource classification suitable for the knowledge graph.

3.3. Schema Layer Design. The architecture of a knowledge graph encompasses both its logical structure and the technical (system) architecture utilized in its construction. From

a logical standpoint, the knowledge graph is bifurcated into two levels as follows: the data layer and the schema layer. In the data layer of the knowledge graph, facts are stored in a graph database. If we consider the fundamental expression of a fact to be in the form of a triplet, either $\langle \text{entity}, \text{relation}, \text{entity} \rangle$ or $\langle \text{entity}, \text{attribute}, \text{entity} \rangle$, then all data stored in the graph database form a vast entity-relationship network, thereby creating a “graph” of knowledge. Situated above the data layer is the schema layer, which constitutes the heart of the knowledge graph. This schema layer houses refined knowledge, typically managed by an ontology library in the context of a knowledge graph. By leveraging the support provided by the ontology library for axioms, rules, and constraints, the schema layer can regulate the relationships between objects such as entities, relationships, entity types, and attributes. The ontology library furnishes a knowledge repository framework with less redundancy than that found in the knowledge graph. In this paper, based on the analysis of the demand for emergency resource knowledge graphs, we constructed the schema layer of the knowledge graph, as depicted in Figure 1.

Definitions for entities and relationships are presented in Tables 1 and 2, respectively. The schema layer comprises eight types of entities. Among these, the primary classification of events (EVE1) represents the four major types of emergencies, i.e., natural disasters, accidents and disasters, public health events, and social security events. The secondary classification of events (EVE2) acts as a sub-classification of the primary classification (EVE1), and the tertiary classification of events (EVE3) serves as a sub-classification of the secondary classification (EVE2). A specific emergency event (EVE) refers to an event that has already occurred and is connected to the tertiary classification of events (EVE3), thereby forming the “related_event” relationship. The specific emergency event (EVE), in conjunction with the time of occurrence (TIME), location of occurrence (LOC), and emergency resources (RES), form the “time_of_occurrence,” “location_of_occurrence,” and “related_resources” relationships, respectively.

When it comes to RES, the variability in quantities associated with source organizations, configuration purposes, and the effects of identical emergency resources in different events can potentially introduce confusion in the knowledge graph’s retrieval logic. To ensure the smooth operation of entity queries and knowledge QA functions, certain entities with complex relationships that are challenging to extract are classified as attributes of a specific entity. Details are shown in Table 3.

3.4. Knowledge Extraction Experiment

3.4.1. Experimental Data and Environment. In this study, the experimental data were sourced from 121 investigation reports related to sudden events, gathered over the past five years from the emergency management departments across various provinces and cities. The data underwent manual labelling to identify entities and relationships, with a focus on five types of entities, i.e., time, location, sudden event,

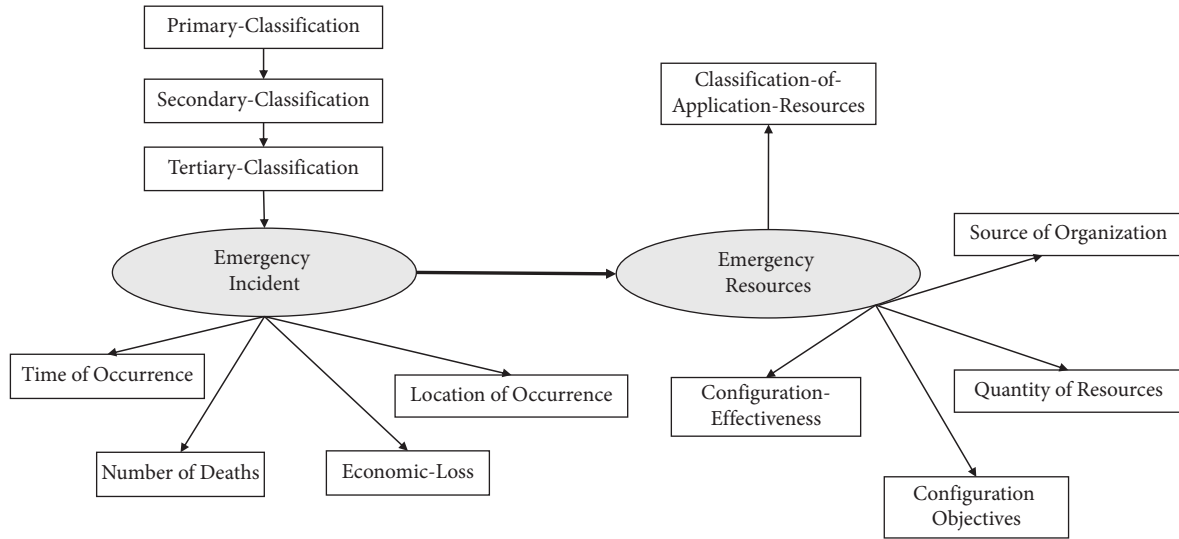


FIGURE 1: Design of the knowledge graph schema layer.

emergency resource, and organization. This was done to facilitate named entity recognition. Following the manual labelling, the labelled entity data were converted into the BIOES format. Any content not classified as an entity was labelled as “O”, while entity “X” was segmented into the beginning, middle, and end parts. The beginning of entity “X” was labelled as “B-X”, the end was labelled as “E-X”, and the remaining parts that were not the start or the end were labelled as “I-X”. For instance, in the phrase “6 heavy excavators,” “B-res” represents the beginning part of the entity, which is an emergency resource. “I-res” is used for the middle part of the entity and “E-res” for the end part. The results of this entity labelling process are depicted in Figure 2.

Herein, the experiments were conducted on a system running Windows 10 with an i5 processor. We used the Paddle and TensorFlow deep learning frameworks to construct the model in a Python environment. Details are shown in Table 4.

3.4.2. Named Entity Recognition Based on BERT + BiLSTM + CRF. The overall architecture of the BERT-BiLSTM-CRF model is depicted in Figure 3. This model primarily consists of three modules. First, the labelled corpus is input into the BERT pretrained language model to generate dynamic word vectors. These vectors are then fed into the BiLSTM module for bidirectional training, which allows for further text feature extraction. Finally, the output from the BiLSTM module is decoded using the conditional random field (CRF) module. This module effectively constrains the dependency relationship between predicted labels to produce a predicted labelling sequence. Various entities in this sequence are then extracted and categorized, yielding the globally optimal sequence, and thus completing the entire process of Chinese entity recognition.

The foremost advantage of this model lies in its application of the BERT language preprocessing model, which eliminates the need for pretraining of character

vectors and word vectors. By directly inputting the sequence into BERT, the model can extract rich word-level syntactic structures and semantic features from the sequence.

For this training, the Adam optimizer was used with a learning rate of 0.001. The LSTM_hidden_size was set to 200; batch_size was 32; epoch was 15; and max_seq_len was set to 256. To prevent overfitting, the dropout technique was used to randomly set input elements to zero, and the parameter value for dropout was set to 0.5. Details are shown in Table 5.

This named entity recognition model was trained using 3463 training, 876 validation data samples, and 289 test samples. The evaluation metrics of the trained model on the test set are as shown in Table 6.

The experimental results clearly demonstrate that the BERT-BiLSTM-CRF model used in this study performs excellently in identifying Chinese-named entities. It outperforms in terms of precision, recall, and F1 score, indicating its superior performance compared to the commonly used BiLSTM-CRF model. Precision P is defined as the ratio of the number of correctly identified entities to the number of identified entities (both correct and incorrect). The recall R is defined as the ratio of the number of correctly identified entities to the total number of entities (both identified and unidentified). The F1 score is defined as the harmonic mean of precision P and recall R. After obtaining the F1 score for each type of entity, we report their unweighted mean as the final macro F1 score. By leveraging the ability of the BERT-BiLSTM-CRF model to capture relationship features between words and contextual nuances, the accuracy of Chinese entity recognition has improved significantly.

After the model training is complete, the model parameters are saved and subsequently loaded for prediction. Once decoding is completed, the predicted result for the sentence, “On December 18th, a gas explosion occurred in

TABLE 1: Entity definitions in the schema layer.

No	Entity category	Entity definition	Example
1	First-level classification of events	EVE1	Natural disasters, accidents
2	Second-level classification of events	EVE2	Meteorological disasters, road traffic accidents, hazardous chemical accidents
3	Third-level classification of events	EVE3	Typhoons, heavy rains, hazardous chemical accidents
4	Specific emergency event	EVE	Major road traffic accident on the "9.28" Wuxi section of the Changshen Expressway, severe "4.30" convective weather disaster
5	Event occurrence time	TIME	January 27, 2022
6	Event occurrence location	LOC	Shengze Town, Wujiang District, Suzhou City, Nanjing
7	Emergency resources	RES	Wind-powered fire extinguishers, disaster relief tents, firefighters
8	Emergency resource type	CLASS	Human resources, engineering equipment supplies, rescue transport supplies

TABLE 2: Relationship definitions in the schema layer.

No	Head entity	Relation definition	Tail entity
1	First-level classification of events	Include	First-level classification of events
2	Second-level classification of events	Include	Third-level classification of events
3	Third-level classification of events	Related_event	Specific emergency event
4	Specific emergency event	Time_of_occurrence	Time of the incident
5	Specific emergency event	Location_of_occurrence	Location of the incident
6	Specific emergency event	Related_resources	Emergency resources
7	Specific emergency event	Secondary_event	Specific emergencies
8	Emergency resources	Belong_to	Emergency resource type

the Yanhu Community of Zhangwan District, Shiyan City. Firefighters have rescued 144 people,” becomes available.

[“B-time”, “I-time”, “I-time”, “I-time”, “I-time”, “E-time”, “O”, “B-loc”, “I-loc”, “I-loc”, “I-loc”, “E-loc”, “O”, “O”, “O”, “O”, “O”, “B-eve”, “I-eve”, “I-eve”, “I-eve”, “E-eve”, “E-eve”, “O”, “B-res”, “I-res”, “I-res”, “E-res”, “O”, “O”, “O”, “O”, “O”, “O”, “O”, “O”]

Finally, using a custom decoding method, the result is as follows:

{“time”: “December 18th”, “loc”: “Yanhu Community, Zhangwan District, Shiyan City”, “eve”: “gas explosion”, “res”: “firefighters”}

3.4.3. Relation Prediction Based on Attention-GRU.

Integrating the attention mechanism with a gated recurrent unit (GRU) model, we have developed an attention-GRU model to predict relationships between entities in the emergency resource field of sudden events, as depicted in Figure 4. The model comprises the following five components: the input vector, GRU hidden layer, attention layer, fully connected layer, and final output. These components process the input vector from the training dataset to produce the predicted output value.

The input vector is introduced into the model via the input layer and subsequently trained. After passing through the GRU hidden layer, the output is a processed vector, which is then utilized as the input for the attention layer. In this attention layer, the weight vector is initially calculated based on the current layer input vector, after which the weight and current layer input vectors are merged to form a new vector. This new vector is then input into the fully connected layer to compute the prediction value.

The GRU and attention layers are the key mechanisms that the model employs to fit the training data. Specifically, the GRU layer retains important information from the input sequence, while the attention layer assigns the learned feature weight to the input vector for the subsequent time step, effectively emphasizing the influence of key features on the load prediction, hence enhancing the prediction performance of the model.

By integrating an attention mechanism, different weights are allocated to various levels of input information. This enhances the impact of key information and ensures that nonessential information is less likely to be extracted.

Consequently, this captures the global relationship between input data and mitigates the problem of information loss caused by sequence length.

The experimental data in this study contained relationships between four different types of entities, with a total of 711 training, 136 validation data samples, and 49 test samples. The relationships are listed in Table 7 and the specific parameters are listed in Table 8.

Upon completion of the model training, the accuracy reached 0.98, and the softmax_loss function value was 2.80992. By introducing entity 1, entity 2, and the prediction statement, the relationship between entities can be determined. In the sentence “At 12:52 on September 5th, a 6.8-magnitude earthquake occurred in Luding County, Ganzi Prefecture, Sichuan Province,” if entity 1 is “the earthquake” (representing an unexpected event) and entity 2 is “Luding County, Ganzi Prefecture, Sichuan Province” (indicating the location where the event occurred), the predicted relationship between these two entities is “location_of_occurrence” (referring to the incident’s location). This prediction carries a probability of approximately 98.2%, as demonstrated in Figure 5.

In a lengthy sentence containing multiple types of entities, the prediction of relationships can also exhibit high accuracy. Consider the sentence, “On August 25th, the 9th typhoon of this year, “Ma’an”, made landfall along the coast of Dianbai, Guangdong. More than 280,000 people, 1,762 pieces of machinery and equipment, over 529,000 woven bags, and over 3.524 million cubic meters of earth and stones have been deployed for inspection and patrol of embankments in the Tarim river basin.” This sentence contains multiple types of entities, such as unexpected events, occurrence time, occurrence location, and emergency resources. If entity 1 is “the typhoon” (an unexpected event) and entity 2 is “the woven bags” (emergency resources), the predicted relationship between these two entities is “configure_resource” (referring to resources configured for the event). This prediction has an approximate probability of 99.2%, as demonstrated in Figure 6.

3.5. Knowledge Graph Querying and Visualization. Data can be integrated into Neo4j through several methods. The LOAD_CSV method allows data import via data import statements in the Neo4j database operation interface.

TABLE 3: Property definitions in the schema layer.

Corresponding entity	No	Attributes	Attributes name	Example
Specific emergencies	1	Casualties	INJ	Four dead, 19 slightly injured
	2	Economic losses	LOSS	Direct economic loss of approximately 3 million yuan
Emergency resources	1	Organization	ORG	Jiangsu Provincial Fire Rescue Corps, Shanghai Maritime Search and Rescue Center
	2	Resource quantity	QUA	More than 35,000 person-times, more than 2,000 tons
	3	Resource allocation purpose	OBJ	Rush to repair and decommission base stations, interrupted optical cables
	4	Resource allocation effect	EFF	Investigate, search, and rescue conducted in the east and west areas of the collapsed building; three people successfully rescued

```

Precipitation has been monitored at 77 regional sites in Yiyang City: B-loc I-loc E-loc 0 0 0
0 0 0 0 0 0 0 0
6 heavy excavators: 0 0 B-res I-res I-res I-res E-res
18th of July: B-time E-time B-time I-time E-time

```

FIGURE 2: BIOES sequence labelling.

TABLE 4: Experimental environment configuration.

Configuration	Parameters
Operating system	Windows 10 64-bit operating system
Programming language	Python 3.7
Deep learning framework	Tensorflow, Paddle
CPU	Intel(R) Core(TM) i5-10210U CPU @ 1.60 GHz 2.11 GHz
Memory	16 GB

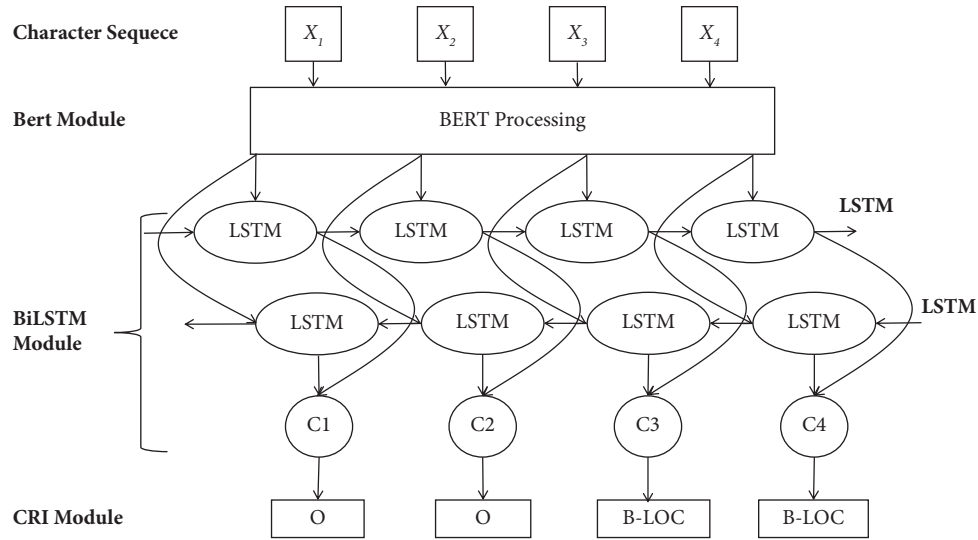


FIGURE 3: BERT-BiLSTM-CRF model framework.

TABLE 5: Specific parameter settings.

Parameters	Value
Optimizer	Adam
Learning rate	0.001
Batch_size	32
Epoch	15
Max_seq_len	256
Lstm_hidden_size	200
Num_layers	2
Drop_out	0.5

However, this method requires data to be converted into the CSV format before importing. Alternatively, the Cypher create statement can be used to import triple data. This is

TABLE 6: Entity extraction results.

Model	Precision (%)	Recall (%)	F1 score (%)
BiLSTM + CRF	89.8	88.8	89.3
BERT + BiLSTM + CRF	89.8	92.7	91.2

typically achieved by connecting to the graph database via the Py2neo library in Python, which has dedicated data import statements. Compared with the latter, the LOAD_CSV method offers a higher data import speed. Hence, this study utilized the LOAD_CSV method for batch data import and the construction of the emergency events' knowledge graph. In the end, 1227 entity nodes, 2080 entity properties, and 2118 entity relationships were established. Following the construction of the emergency resource

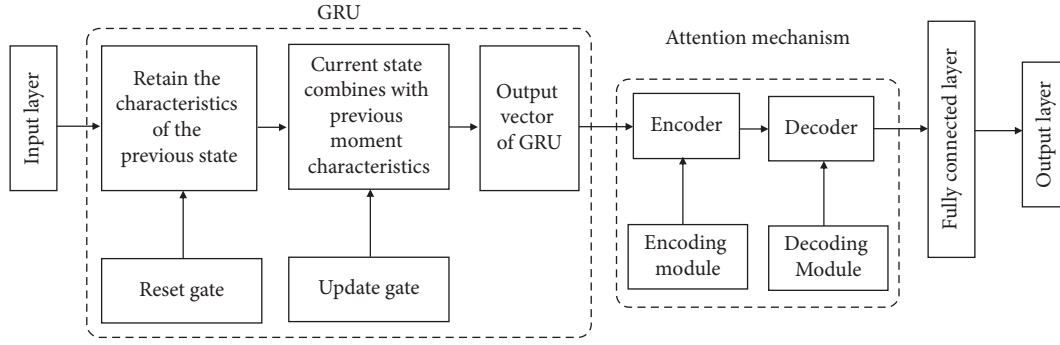


FIGURE 4: Attention-GRU model framework.

TABLE 7: Entities and relationships related to relationship prediction.

	Entity 1	Relationship	Entity 2
1	Sudden event	Time_of_occurrence	Occurrence time
2	Occurrence time	Location_of_occurrence	Occurrence location
3	Sudden event	Configure_resource	Emergency resource
4	Emergency resource	Resource_source	Organization

TABLE 8: Attention-GRU model parameter settings.

Parameter entity	Value
Gru_size	230
Keep_prob	0.5
Num_layers	1
Num_classes	4
Num_epoch	80
Num_step	70
Pos_size	5
Vocab_size	16691

knowledge graph, knowledge queries were performed through Cypher in the Neo4j graph database. The query intents and statements are provided in Table 9, and the corresponding results are depicted in Figures 7–9.

Through the knowledge graph of emergency resources for sudden events, relevant departments can learn various information about sudden events and emergency resources. Thus, knowledge graphs provide valuable references for summaries in the generation of responses to sudden events and formulation of emergency plans. For example, querying all classifications of natural disasters is illustrated in Figure 7, specific events of accidents and disasters are shown in Figure 8, and classification of relevant emergency resources involved in a certain sudden event is shown in Figure 9.

4. Design and Implementation of the Emergency Resource Knowledge Graph QA System

Intelligent QA is a representative application of artificial intelligence. While most people obtain knowledge via search engines, these general search engines often suffer from a lack of specificity, producing numerous disordered results in specific professional fields. As a consequence, the search

results often do not meet users' expectations. QA systems, on the other hand, can provide users with convenient and effective knowledge services, helping them obtain the targeted answers they seek. The emergency resource knowledge graph constructed in this study encompasses a large number of emergency cases and diverse emergency resource information, thus supporting the implementation of intelligent QA functions. The architecture of the knowledge QA system is depicted in Figure 10.

4.1. Knowledge Question Design. Research on intelligent QA systems related to emergency events and emergency resource knowledge graphs is limited in China. Furthermore, there are currently no open-source datasets that can be used directly. Therefore, based on the existing data information in the constructed knowledge graph of emergency events and resources, this study designed the following three categories of question queries: emergency events, emergency resources, and comprehensive categories. These categories were further subdivided, resulting in a total of 14 distinct knowledge question query intents, as detailed in Table 10.

4.2. Knowledge QA Method Design. When performing analysis of user queries, the relatively limited scope of existing data on emergency resource knowledge questions and answers allows for ease in manually summarizing common types of questions. This ensures the accuracy of the responses by designing knowledge query templates for each question type. Therefore, this study adopted a template rule-based approach for question analysis. After applying the question classification algorithm to the user's query, the user's search intent could be identified, and the corresponding knowledge query template was used to provide the answer. The process is shown in Figure 11.

Please enter a sentence, Format is 'name1 name2 sentence' : *earthquake Luding County, Garze Prefecture, Sichuan Province On September 5th at 12:52, a 6.8 magnitude earthquake occurred in Luding County, Garze Prefecture, Sichuan Province.*

entity1 : earthquake
entity2 : Luding County, Garze Prefecture, Sichuan Province
On September 5th at 12:52, a 6.8 magnitude earthquake occurred in Luding County, Garze Prefecture, Sichuan Province.
Relation is :
No. 1 : Location_of _occurrence, Probability is 0.9823596
No. 2 : Time_of occurrence, Probability is 0.016907735
No. 3 : Configure_resource, Probability is 0.00046967473

FIGURE 5: Short sentence relationship prediction.

Please enter a sentence, Format is 'name1 name2 sentence' : *typhoon woven bags On August 25th, Typhoon Ma'ang, the 9th typhoon of this year, made landfall along the coast of Dianbai, Guangdong. More than 280000 people, 1762 machinery and equipment, more than 529000 woven bags, and more than 3.524 million cubic meters of earthwork were deployed in the Tarim River Basin*

entity1 : typhoon
entity2 : woven bags
On August 25th, Typhoon Ma'ang, the 9th typhoon of this year, made landfall along the coast of Dianbai, Guangdong. More than 280000 people, 1762 machinery and equipment, more than 529000 woven bags, and more than 3.524 million cubic meters of earthwork were deployed in the Tarim River Basin.
Relation is :
No. 1 : Configure_resource, Probability is 0.992585
No. 2 : Time_of occurrence, Probability is 0.0044843256
No. 3 : Resource_Quantity, Probability is 0.0024247007

FIGURE 6: Long-sentence relationship prediction.

4.2.1. Preprocessing of Questions. When a user submits a question via the QA interface, the question undergoes pre-processing and then entity recognition is performed to identify the best-matching QA template. In this study, we utilized Jieba, a Chinese text segmentation library. The tokenizer first removes blank characters and punctuation marks from the question, resulting in a cleansed question consisting solely of Chinese characters. Jieba then performs part-of-speech tagging and word segmentation. To enhance the precision of the word segmentation results, we created an emergency resource knowledge dictionary. The entity and attribute information from the previously constructed emergency resource knowledge graph was stored in a text file to create this dictionary. Information regarding part-of-speech entities is presented in Table 11.

4.2.2. Question Template Definition. Given the 14 types of knowledge QA query intents mentioned above, corresponding question templates were designed. These

templates fall into two categories as follows: entity query templates and attribute query templates. When the user's natural language question fits the rules of a template, a corresponding question template is matched. Each question template aligns with a database query statement, and the query can be resolved after populating specific data such as entities, attributes, and categories. Table 12 outlines the query statements corresponding to each template.

4.2.3. Template Mapping and Knowledge Retrieval. The template mapping operation is performed by the template mapping function, which takes the problem preprocessing and classification results as parameters. This function reads the Cypher statement template corresponding to the problem and replaces the tags in the template with the entity information from the question to form the final Cypher query statement.

TABLE 9: Knowledge graph query cases.

Query intention	Query statement
The first, second, and third level classifications of natural disasters	Match data = (na:EVE1{name: "Natural disasters."}) [re]->(nb:EVE2)-[re2]->(nc:EVE3) return data
Specific events of accident disasters	Match data = (na:EVE1{name: "Natural disasters."})-[re]->(nb:EVE2)-[re2]->(nc:EVE3)-[re3]->(nd:EVE) return data limit 40
Classification of emergency resources related to a certain emergency event	Match data = (n:EVE{name: "xxx"})-[re]->(name:RES)-[re2]->(name:CLASS) return data

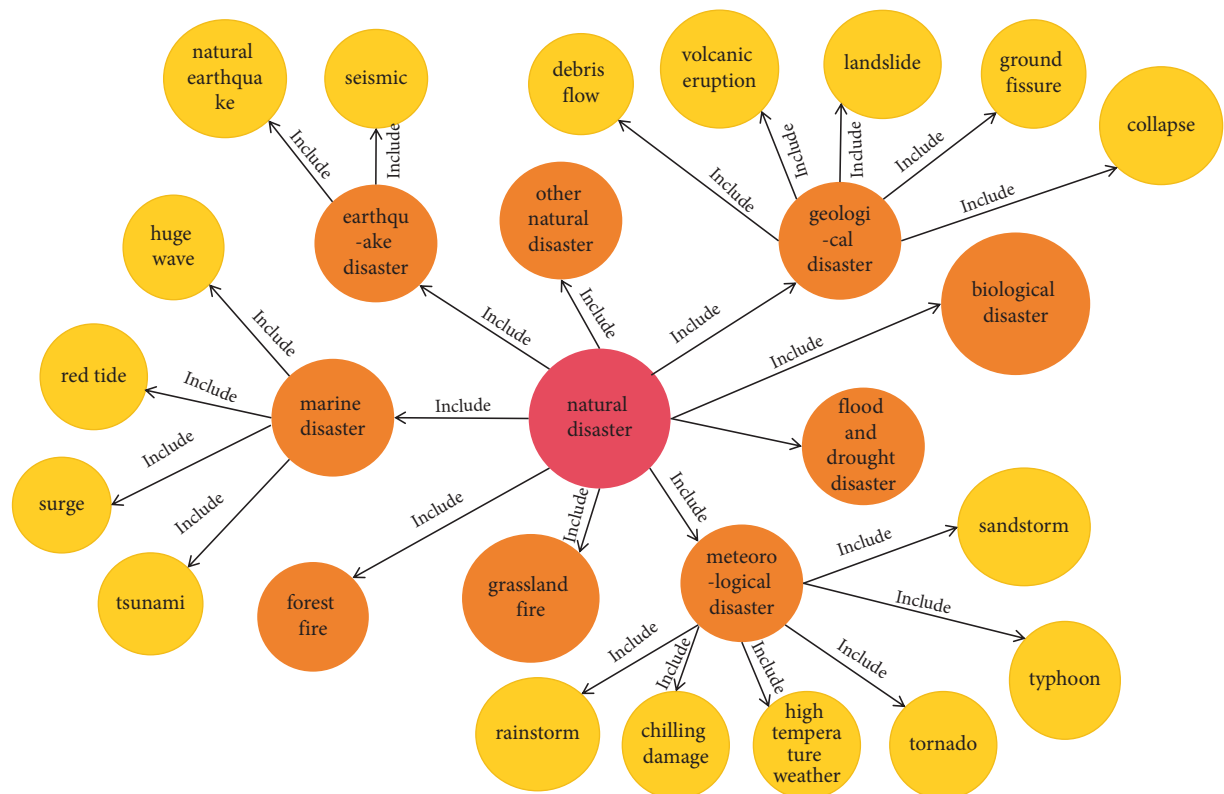


FIGURE 7: Classification of natural disasters at first, second, and third levels.

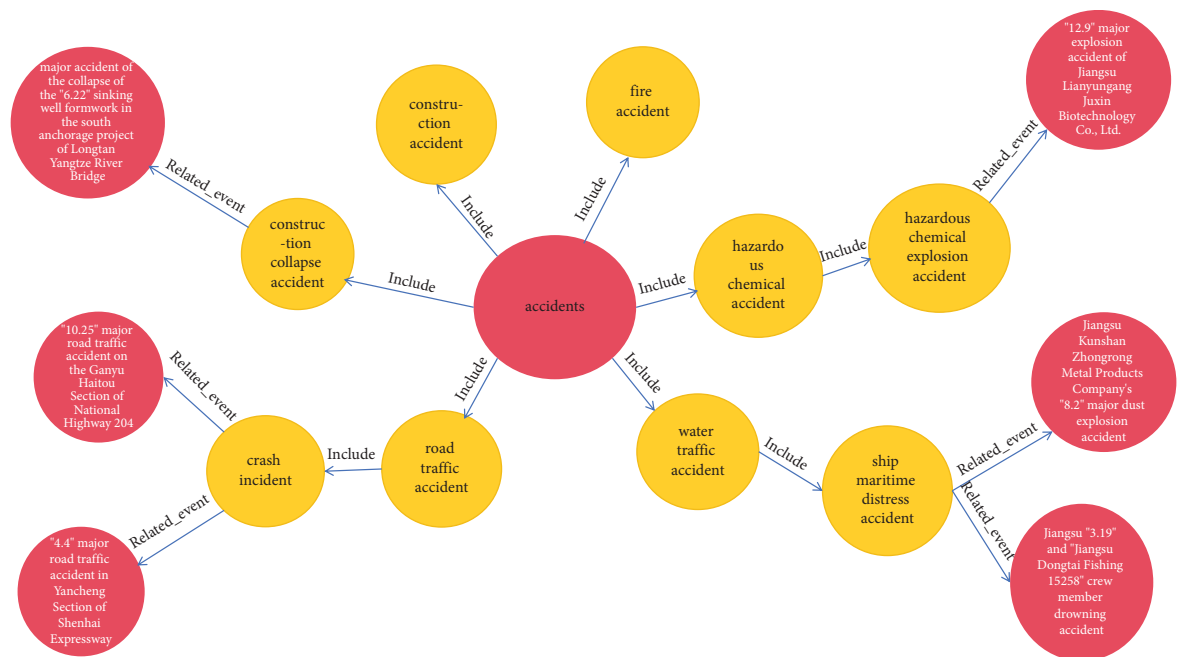


FIGURE 8: Specific events of accidents and disasters.

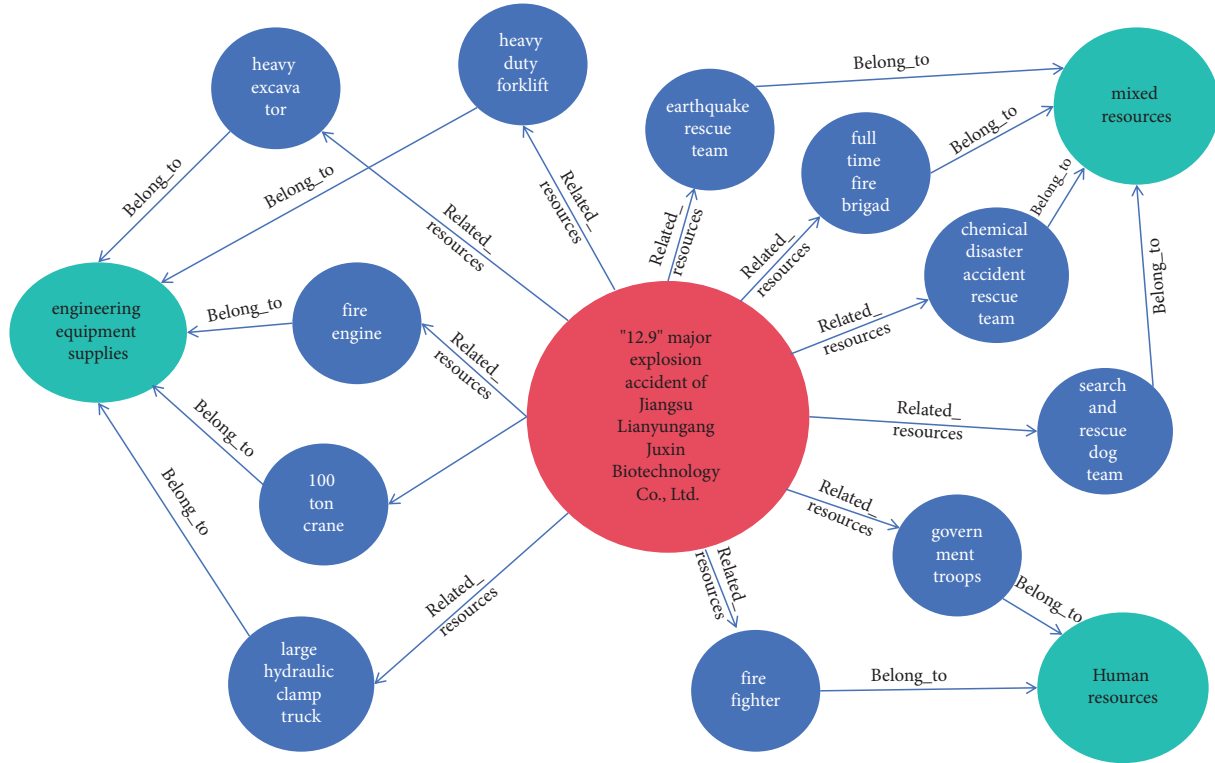


FIGURE 9: Classification of resources for the “12.9” major explosion accident of Jiangsu Lianyungang Juxin Biotechnology Co., Ltd.

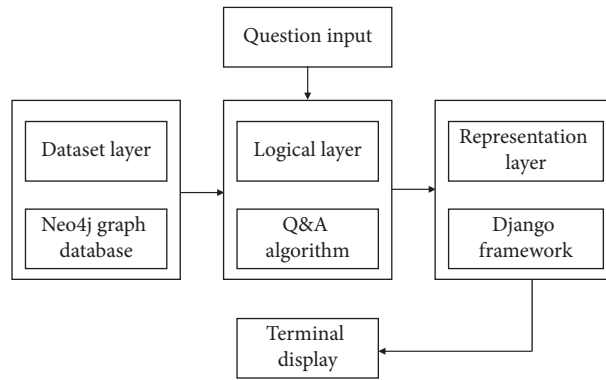


FIGURE 10: Frame diagram of the knowledge QA system.

For instance, consider the natural language question, “what emergency resources are involved in the “4.30” strong convective weather disaster?” For this type of query, the template mapping function first extracts the emergency entity, EVE, from the user’s question after word segmentation: ““4•30” strong convective weather disaster.” The query “emergency resources” is classified as entity question template 4, according to the presence of EVE entity + “emergency resources” in the query, i.e., “emergency resources involved in an emergency.”

Subsequently, the corresponding Cypher query template is applied by passing the emergency entity into the query template to generate the final query statement. The Cypher statement template for this type of query is “MATCH (n:EVE)-[r:“Related

Resource”]->(b:RES), where n.name = “EVE_name” return b.” After passing the incident entity into the template, the final query statement becomes “MATCH (t: Time)-[r: time]->(a: Accident) where t.name = “4•30” strong convective weather disaster “return b.””

The generated Cypher statement is executed to retrieve the corresponding answer from the Neo4j graph database. The result is returned to the user after a simple formatting. The background processing results of template mapping and knowledge retrieval are shown in Figure 12.

4.3. System Display. The visualization part of this QA system is based on the Django framework and ECharts components are displayed in different forms according to the different

TABLE 10: Design of knowledge quiz questions.

Issue classification	Serial Number	Query intent	Example of a problem
Emergency-type issues	1	Emergencies that occur at a certain time (first-level classification)	What natural disaster events occurred in 2022?
	2	Emergencies that occur at a certain time (secondary classification)	What meteorological disaster events occurred in 2022?
	3	Emergencies that occur at a certain time (tertiary classification)	What rainstorm events occurred in 2022?
	4	Emergencies that occur at a certain time and place	What emergencies occurred in Nanjing in 2022?
	5	The number of casualties in an emergency	What are the casualties caused by the “4.30” strong convective weather disaster?
	6	The economic loss of an unexpected event	What is the economic loss caused by the “4.30” strong convective weather disaster?
	7	A secondary event of an emergency	What are the secondary events of the “4.30” strong convective weather disaster?
Emergency resource class issues	8	Emergency resources under a certain type of emergency resource	What are the engineering equipment materials?
	9	The type of emergency resource	What type of emergency resources does an ambulance belong to?
Comprehensive questions	10	Emergency resources involved in an emergency	What emergency resources were involved in the “8.28” explosion accident in Jiangyin City?
	11	The number of emergency resources in an emergency	How many snow removal personnel were allocated in Jiangsu province during the “2.7” medium to heavy snowfall event?
	12	The source of an emergency resource in an emergency	What organization did the snow removal personnel come from in the “2.7” medium to heavy snow event in Jiangsu province?
	13	The effect of an emergency resource allocation in an emergency	In the “8.28” explosion accident in Jiangyin City, what effect did the life detector play?
	14	The purpose of an emergency resource allocation in an emergency	What was the purpose of the emergency rescue team for the “4.30” strong convective weather disaster?

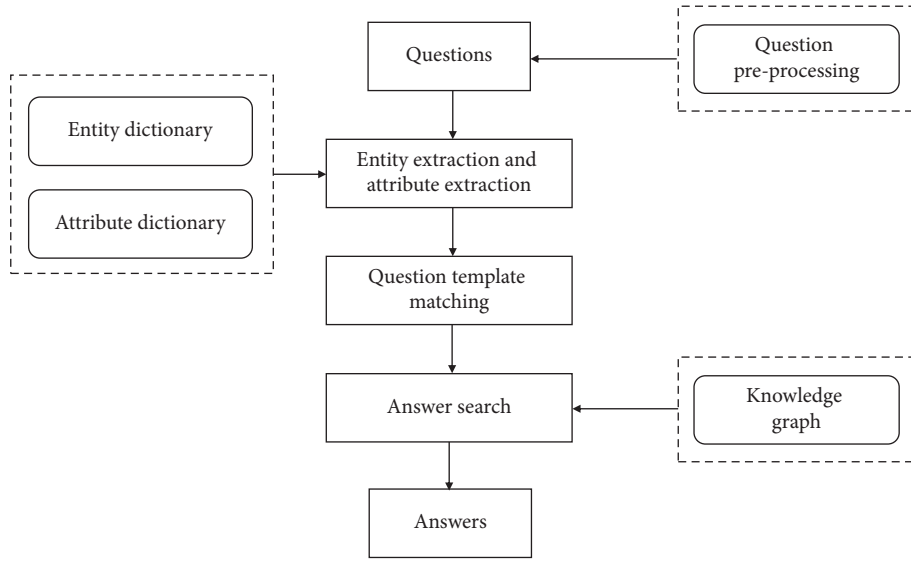


FIGURE 11: Problem-matching flowchart.

TABLE 11: Entity part-of-speech table.

Part of speech	Meaning
EVEC	Incident classification
TIME	Time
PLACE	Place
EVE	Specific emergencies
NOTHING	Emergency resources
CLASS	Emergency resource classification

types of questions entered by the user. The interface of this system is shown in Figure 13. The existing functions of the system include map query display and knowledge QA.

5. Summary

Emergencies occur frequently in China, and the type of emergency determines the required emergency resources. Appropriate allocation of these resources can significantly mitigate the losses caused by emergencies. The proposed knowledge graph can effectively organize and represent the information on emergency resources and solve the problem of unclear feature input in emergency resources tasks. Using the Neo4j graph database, this study constructed a knowledge graph of emergency resources.

An analysis of emergency and emergency resource classification allowed us to define a more detailed classification of emergency resources. We constructed a model layer of the emergency resource knowledge graph, which includes eight types of entities—first-level classification, second-level classification, tertiary classification, specific emergency, event time, incident location, emergency resource, and emergency resource type—and eight types of

relationships between these entities. Specific emergencies had attributes such as the number of casualties and economic losses, while the emergency resources had attributes like the number of resource allocations, resource source organization, resource allocation purpose, and resource allocation effect.

Knowledge extraction from the information provided by the Ministry of Emergency Management and provincial and municipal emergency investigation reports was achieved through manual marking, allowing us to design an extraction scheme for entities and relationships. This scheme included named entity recognition based on the Bert-BiLSTM-CRF model and relationship prediction based on the attention-GRU model. Following data extraction and integration, the Neo4j graph database stored the knowledge graph of emergency resources as attribute graphs, addressing the limitations of relational databases in knowledge storage and construction of emergency knowledge graphs.

After the construction of the knowledge graph, we designed a total of 14 types of problem categories in the emergency resources field, based on the existing data types in the knowledge graph and the built-in template QA method. We constructed the Cypher query template corresponding to each problem. The tokenizer tagged and split parts of speech in the questions, aligning the entity in the question with the emergency resource knowledge dictionary. This process helped us find the corresponding question template through the template mapping function, replace the question entity with the label in the Cypher statement, and query the answer, thereby realizing the domain QA function.

Finally, using Django as the web framework, we designed an emergency resource QA system, effectively showcasing the functions of graph query and intelligent knowledge QA for knowledge graph applications.

TABLE 12: Classification of problem templates.

Issue template	Serial number	Query intent
Entity query template	1	Emergencies that occur at a certain time (first-level classification)
	2	Emergencies that occur at a certain time (secondary classification)
	3	Emergencies that occur at a certain time (tertiary classification)
	4	Emergency resources involved in an emergency
	5	Emergencies that occur at a certain time and place
	6	Emergency resources under a type of emergency resource
	7	The type of emergency resource
	8	Secondary event of an emergency
	9	Determine the category to which the emergency belongs
	10	Determine the classification to which the emergency resource belongs
	11	Determine whether an emergency involves an emergency resource
First property queries template	1	Number of casualties in an emergency
	2	Economic loss of an unexpected event
	3	Number of emergency resources in an emergency
	4	Source of an emergency resource in an emergency
	5	Purpose of emergency resource allocation in an emergency
	6	Effect of emergency resource allocation in an emergency
	7	Comparison of economic losses in emergencies
	8	Comparison of the number of the same emergency resources in different emergencies

====

['What', 'emergency resources', 'are involved', 'in', 'the 430 severe convective weather disaster', '?']
 {'EVE': 'the 430 severe convective weather disaster'}
 Hit entity template 4
 MATCH (n:EVE)-[r:'Related_resources']->(b:RES) where n.name='the 430 severe convective weather disaster' RETURN b
 (_359:RES {EFF:' ', OBJ: '\u62A2\u4fee\u9000\u670d\u57fa\u7ad9\u3001\u4e2d\u65ad\u5149\u7f06',
 ORG: '\u7701\u79fb\u52a8\u3001\u7535\u516c\u53f8', QUA: '2601\u8f86\u6b21',
 name: '\u901a\u4fe1\u4fdd\u969c\u8f66\u8f86'})
 Found node: {'EFF': ' ', 'ORG': 'Provincial Mobile, Telecom, China Unicom Companies', 'OBJ': 'Emergency repair and service
 withdrawal of base stations, interruption of optical cables', 'name': 'Communication support vehicle', 'QUA': '2601 vehicles'}
 (_358:RES {EFF:' ', OBJ: '\u62A2\u4fee\u9000\u670d\u57fa\u7ad9\u3001\u4e2d\u65ad\u5149\u7f06',
 ORG: '\u7701\u79fb\u52a8\u3001\u7535\u516c\u53f8',

FIGURE 12: Template mapping and knowledge retrieval results.

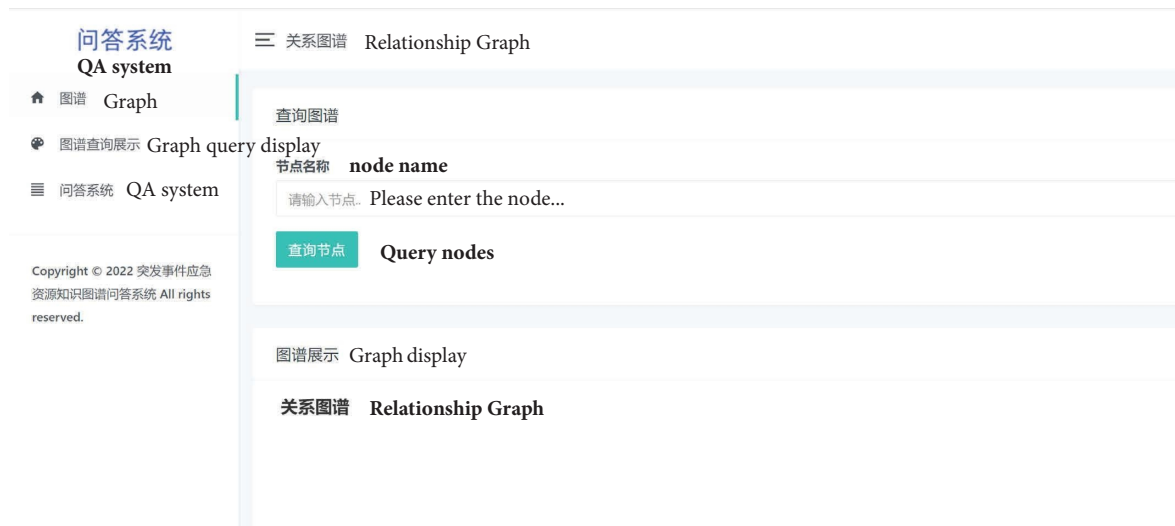


FIGURE 13: Display of the main interface of the QA system.

Data Availability

The data used to support the findings of this study are available on request from the corresponding author.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (project numbers are 72274096, 71774084, 72301136, and 72174087), the Foreign Cultural and Educational Expert Program of the Ministry of Science and Technology of China (G2022182009L).

References

- [1] X. Zhou, "A comprehensive framework for assessing navigation risk and deploying maritime emergency resources in the South China Sea," *Ocean Engineering*, vol. 248, Article ID 110797, 2022.
- [2] L. Chen, Y. Li, Y. Chen, N. Liu, C. Li, and H. Zhang, "Emergency resources scheduling in distribution system: from cyber-physical-social system perspective," *Electric Power Systems Research*, vol. 210, Article ID 108114, 2022.
- [3] Y. Du, H. Xiao, J. Sun et al., "Hierarchical pre-positioning of emergency resources for a chemical industrial parks concentrated area," *Journal of Loss Prevention in the Process Industries*, vol. 66, Article ID 104130, 2020.
- [4] Q. Ma, D. Zhang, C. Wan, J. Zhang, and N. Lyu, "Multi-objective emergency resources allocation optimization for maritime search and rescue considering accident black-spots," *Ocean Engineering*, vol. 261, Article ID 112178, 2022.
- [5] Y. P. Meng, "The optimization and coordination of multi-layer recruitment network for social emergency resources," *Computers and Industrial Engineering*, vol. 130, pp. 50–61, 2019.
- [6] D. Chen, X. Fang, Y. Li, S. Ni, Q. Zhang, and C. K. Sang, "Three-level multimodal transportation network for cross-regional emergency resources dispatch under demand and route reliability," *Reliability Engineering and System Safety*, vol. 222, Article ID 108461, 2022.
- [7] Z. Liao, B. Wang, X. Xia, and P. M. Hannam, "Environmental emergency decision support system based on Artificial Neural Network," *Safety Science*, vol. 50, no. 1, pp. 150–163, 2012.
- [8] X. Yin, X. Xu, and B. Pan, "Selection of Strategy for Large Group Emergency Decision-making based on Risk Measurement," *Reliability Engineering and System Safety*, vol. 208, 2021.
- [9] Y. Chen, L. Zhang, J. Hu, C. Chen, X. Fan, and X. Li, "An emergency task recommendation model of long-distance oil and gas pipeline based on knowledge graph convolution network," *Process Safety and Environmental Protection*, vol. 167, pp. 651–661, 2022.
- [10] A. Hogan, E. Blomqvist, M. Cochez et al., "Knowledge graphs," *ACM Computing Surveys*, vol. 54, no. 4, pp. 1–37, 2021.
- [11] N. Xiao, B. Peng, X. Li, J. Wu, J. Lou, and Y. Si, "Research on the construction and implementation of power grid fault handling knowledge graphs," *Energy Reports*, vol. 9, pp. 182–189, 2023.
- [12] E. Shin, S. Yoo, Y. Ju, and D. Shin, "Knowledge graph embedding and reasoning for real-time analytics support of chemical diagnosis from exposure symptoms," *Process Safety and Environmental Protection*, vol. 157, pp. 92–105, 2022.
- [13] S. Mao, Y. Zhao, J. Chen, B. Wang, and Y. Tang, "Development of process safety knowledge graph: a Case study on delayed coking process," *Computers and Chemical Engineering*, vol. 143, Article ID 107094, 2020.
- [14] Y. Zhang and J. Yang, "Chinese ner using lattice lstm," 2018, <https://arxiv.org/abs/1805.02023>.
- [15] X. Li, H. Yan, X. Qiu, and X. Huang, "Flat: chinese ner using flat-lattice transformer," 2020, <https://arxiv.org/abs/2004.11795>.
- [16] B. Jehangir, S. Radhakrishnan, and R. Agarwal, "A survey on Named Entity Recognition—datasets, tools, and methodologies," *Natural Language Processing Journal*, vol. 3, Article ID 100017, 2023.
- [17] X. Guo, H. Zhang, H. Yang, L. Xu, and Z. Ye, "A single attention-based combination of CNN and RNN for relation classification," *IEEE Access*, vol. 7, pp. 12467–12475, 2019.
- [18] T. Gan, Y. Gan, and Y. He, "Subsequence-level entity attention lstm for relation extraction," in *Proceedings of the 2019 16th International Computer Conference on Wavelet Active Media Technology and Information Processing*, Chengdu, China, December 2019.
- [19] A. Katiyar and C. Cardie, "Going out on a limb: joint extraction of entity mentions and relations without dependency trees," in *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Vancouver, Canada, July 2017.
- [20] Y. Liu, A. Li, J. Huang et al., "Joint extraction of entities and relations based on multi-label classification," in *Proceedings of the 2019 IEEE Fourth International Conference on Data Science in Cyberspace (DSC)*, Hangzhou, China, June 2019.
- [21] S. Behmanesh, A. Talebpour, M. Shamsfard, and M. M. Jafari, "Improved relation span detection in question answering systems over extracted knowledge bases," *Expert Systems with Applications*, vol. 224, Article ID 119973, 2023.
- [22] A. M. Turing, *Computing Machinery and Intelligence*, Springer, Berlin, Germany, 2009.
- [23] C. Liu, X. Ji, Y. Dong, M. He, M. Yang, and Y. Wang, "Chinese mineral question and answering system based on knowledge graph," *Expert Systems with Applications*, vol. 231, Article ID 120841, 2023.
- [24] X. Wang, Z. Luo, R. He, and Y. Shao, "Novel medical question and answer system: graph convolutional neural network based with knowledge graph optimization," *Expert Systems with Applications*, vol. 227, Article ID 120211, 2023.
- [25] A. Li, Q. Wei, C. Han, and X. Xing, "Research on the construction of smart care question answering system based on knowledge graph," *Procedia Computer Science*, vol. 214, pp. 1595–1602, 2022.
- [26] A. Jaya Kumar, C. Schmidt, and J. Köhler, "A knowledge graph based speech interface for question answering systems," *Speech Communication*, vol. 92, pp. 1–12, 2017.